

CAP stands for **Consistency, Availability, and Partition Tolerance**, and the theorem states that:

It is impossible for a distributed data store to simultaneously provide all three guarantees.

- **Consistency (C)**: Every read receives the most recent write or an error.
- **Availability (A)**: Every request (read or write) receives a non-error response, without guarantee that it contains the most recent write.
- **Partition Tolerance (P)**: The system continues to operate despite an arbitrary number of messages being dropped (or delayed) by the network between nodes.

Three Pillars of CAP

1. Consistency

Consistency ensures that **every read receives the most recent write or an error**. This means that all working nodes in a distributed system will return the same data at any given time.

In a consistent distributed system, if you write data to node A, a read operation from node B will immediately reflect the write operation on node A.

Consistency is crucial for applications where having the most up-to-date data is critical, such as financial systems, where a balance inquiry must reflect the most up-to-date state of an account.

2. Availability

Availability guarantees that **every request (read or write) receives a response**, without ensuring that it contains the most recent write. This means that the system remains **operational** and **responsive**, even if the response from some of the nodes don't reflect most up-to-date data.

Availability is important for applications that need to remain operational at all times, such as online retail systems.

3. Partition Tolerance

Partition Tolerance means that the **system continues to function despite network partitions** where nodes cannot communicate with each other.

A **network partition** occurs when a network failure causes a distributed system to split into two or more groups of nodes that cannot communicate with each other.

When there is a network partition, the system must choose between **Consistency** and **Availability**.

Partition Tolerance is essential for distributed systems because network failures can and do happen. A system that tolerates partitions can maintain operations across different network segments.